

Circularity for a Sustainable Future

Class description

The Circular Economy seeks to redefine growth and development, decoupling it from finite resource consumption and eliminating waste from the system. Underpinned by a transition to renewable energy and materials, it generates economic, social and environmental capital based on three principles: i) eliminate waste and pollution; ii) circulate products and materials at their highest value for longer; and iii) regenerate nature.

Circularity for a Sustainable Future seeks to promote sustainable and inclusive entrepreneurship initiatives among students. It will familiarize them with the Circular Economy fundamentals: its principles, key concepts and most common tools and business models, combined with Human-Centered Design principles and methodology to equip students with the tools necessary to design and implement circular and inclusive projects aimed at tackling the most pressing social, environmental and economic problems.

Class objectives

At the end of the course, students will be able to:

1. Think critically, systemically, and circularly;
2. Fluently speak the Circular Economy language; and
3. Identify, design and value circular solutions in any sector, from an economic, social, and environmental point of view.

Teaching method

The course will be taught mainly through a combination of the lecture, discussion, and assignment-based methods. In addition to quizzes and exams, students will work on four types of complementary activities throughout the course:

1. Prior to each class: reading of relevant reports and success stories;
2. During each class: discussion and short exercises;
3. Throughout the course, individually: research and formulation of a specific project for the chosen sector; and
4. Throughout the course, in a group: idea exchange and comparison of differences and similarities (to be shared with the class in the form of a sectorial presentation).

Student performance evaluation throughout the course is presented in Table 1, while the course work program is presented in Table 2.

Target audience

This course can be taken as an optional subject for undergraduate degrees in Business, Economics, Political Science, International Relations and related subjects. Previous experience with the topics covered is not required, but a basic understanding of business, management or public policy would be helpful.

Table 1. Evaluation

Element	Description	Grade
Class participation	Quality of student interventions: questions, comments, examples, etc.,	5%
Quizzes	Five quizzes to verify the understanding of basic concepts.	10%
Group presentations	Three group presentations summarizing sectoral needs, differences and similarities between projects/ programs, and common challenges and opportunities identified.	15%
Partial exam	One exam with three short essay questions, covering the topics discussed in classes 1 to 10.	15%
Individual partial deliveries	In preparation for the final project, students will submit two partial projects of two pages each: 1. General context of the sector and location they have chosen and the problem they want to tackle, highlighting the importance of solving it and the approach they'd like to work from (public, private or civil society) and why; and 2. Profile of their project or program, highlighting the business/ operational model, product or service, and target customer/ beneficiary group.	15%
Final exam	One exam with multiple choice, true-false and completion questions, and one essay question, covering the topics discussed in classes 1 to 14.	15%
Final report	Detailed report of the chosen sector and location, legal and institutional framework within which the proposed project or program would be operating, description of the problem or need to address, implementation approach, business/ operational model, product/ service, target customer/ beneficiary group, expected challenges and opportunities, and basic financial model and evaluation framework.	25%

Course bibliography

- Stroh, David Peter,, *Systems Thinking for Social Change* (2015)
- Webster, Ken and Ellen MacArthur, *The Circular Economy: A Wealth of Flows* (2015)
- Lacy, Long, et al., *The Circular Economy Handbook: Realizing the Circular Advantage* (2020)

Table 2. Course work program

Module	Class	Suggested materials to review before class	Disussion guide	Other
Intro	1: Sustainability and the 2030 Agenda for Sustainable Development	- Class syllabus - Kim, Daniel H., <i>Introduction to Systems Thinking</i> - UK Government Office for Science, <i>The Futures Toolkit</i> - UN, <i>Transforming our World: The 2030 Agenda for Sustainable Development</i> - IGES, <i>SDG Interlinkages and Network Analysis</i>	- Schools of thought and theory, and basic concepts - Systems, futures and design thinking 2030 Agenda: background and structure, SDGs, mainstreaming and evaluation	
	2: The Fourth Industrial Revolution and Introduction to the Circular Economy	https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-are-industry-4-0-the-fourth-industrial-revolution-and-4ir https://ellenmacarthurfoundation.org/videos/basics-of-a-circular-economy https://assets.website-files.com/5e185aa4d27bcf348400ed82/63ecb3ad94e12d3e5599cf54_CGR%202023%20-%20Report.pdf	4.0 technologies, technology adoption curve and innovation matrix - Material flows and life-cycle analysis - Strategies and business models - Social considerations of technological disruption	
	3: Introduction to Human-Centered Design and the Circular Business Canvas	https://www.youtube.com/watch?v=ldYzbV0NDp8 - Hasso Plattner Institute of Design at Stanford, <i>An Introduction to Design Thinking Process Guide</i> https://www.circulardesignguide.com/post/circular-business-model-canvas	- HCD: Background, principles and mindset - HCD: Process stages - CBC: what it is and how it differs from other tools	Quiz #1
	4: Diversity, Equity and Inclusion	http://dx.doi.org/10.17502/m.rcs.v6i2.233 https://www.unwomen.org/sites/default/files/2022-10/Progress-on-the-sustainable-development-goals-the-gender-snapshot-2022-es.pdf http://doi.org/10.29101/crcs.v29i0.17792	- What it is and why it's a pillar of sustainability - Frameworks and debates - Intersectionality	Marisol Aguilar/ Ángeles Cruz (RacismoMX)
	5: Group presentation #1	Other groups' presentations, prepare 3 questions for each	- Chosen sector and justification - Legal and institutional framework - Problems identified and why it's important to solve them	Quiz #2 Partial individual report #1

Module	Class	Suggested materials to review before class	Disussion guide	Other
Sectorial deep-dive	6: Plastics	https://www.sciencehistory.org/sites/default/files/history-of-plastics.pdf https://theconversation.com/when-biodegradable-plastic-is-not-biodegradable-116368 https://ellenmacarthurfoundation.org/circular-examples/an-industry-wide-shared-packaging-system-swedish-return-system	- Brief history and typology of plastics - Alternatives and treatments - INC on Plastic Pollution and the New Plastics Economy	
	7: Textiles	https://www.forbes.com/sites/forbesagencycouncil/2020/12/01/circular-economy-and-the-textile-industry/?sh=7ddd261157a9 https://pacecircular.org/sites/default/files/2021-02/circular-economy-action-agenda-textiles.pdf https://ellenmacarthurfoundation.org/circular-examples/hm-group	- Why textiles? - Alternatives and treatments - What's already under way	
	8: Food and agriculture	https://www.unwater.org/water-facts/water-food-and-energy https://council.science/wp-content/uploads/2017/03/SDGs-interactions-2-zero-hunger.pdf https://www.circle-economy.com/circular-food-systems https://ellenmacarthurfoundation.org/circular-examples/closing-the-nutrient-loop	- SDG#2 interlinkages and WFE-CC Nexus - Alternatives	Quiz #3
	9: Construction and circular cities	https://ellenmacarthurfoundation.org/topics/cities/overview https://circularcitiesdeclaration.eu https://ellenmacarthurfoundation.org/circular-examples/regional-programme-for-a-circular-economy-brussels	- The built environment - What is a circular city - Frameworks and evaluation	
	10: Group presentation #2	Other groups' presentations, prepare 3 questions for each	- Target customer group - Chosen products or services and possible strategies	Quiz #4 Partial individual report #2
	11: Partial exam	Classes 1-10.		

Module	Class	Suggested materials to review before class	Disussion guide	Other
Circular project design	12: Material flows, life-cycle analysis and life-cycle redefinition	• https://www.greenbiz.com/article/extending-product-life-build-circular-economy • https://ellenmacarthurfoundation.org/circular-economy-diagram • https://www.bpf.co.uk/sustainable_manufacturing/life-cycle-analysis-lca.aspx	• Butterfly diagram • Life-cycle analysis fundamentals • CBC: Customer segments, value proposition and channels	
	13: Circular strategies and business models	• https://pre-sustainability.com/articles/5-roads-to-circular-economy-part-i-product-life-extension/ • https://www.tandfonline.com/doi/epdf/10.1080/09537287.2018.1449247?needAccess=true&role=button • https://www.boardofinnovation.com/circular-economy-business-models-explained/#:~:text=Circular%20business%20model%20definition,minimizing%20ecological%20and%20social%20costs	• 10 Rs • Design, optimal use, value recovery, and circular support models • CBC: Customer relationships, revenue streams and key resources	
	14: Evaluating circularity	• https://ellenmacarthurfoundation.org/resources/circulytics/overview • https://ec.europa.eu/newsroom/env/items/624232/default • https://www.bsigroup.com/en-GB/standards/benefits-of-using-standards/becoming-more-sustainable-with-standards/Circular-Economy/BS-8001---Circular-economy/	• Linear vs circular evaluation • Circular evaluation frameworks • CBC: Key activities, key partners and costs	
	15: Group presentation #3	• Other groups' presentations, prepare 3 questions for each	• Business models • Expected challenges and opportunities	Quiz #5 Final report delivery.
	16: Final exam	• Classes 1-15.		
	17: Conclusion	• None	• General feedback on final reports • Course highlights and takeaways • Suggestions on course content and format	